Emergent Gravity from the Φ_{MDL} Field: Einstein Equations, Kink Sources, and Quantum Gravity Scale

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Abstract

Starting from the $\mathbb{Z}_7$-symmetric Klein–Gordon field Φ_{MDL} with $V = (m_\varphi^2/49)(1 - \cos 7\Phi)$ and $m_\varphi = m_\tau = 1776.86\,\text{MeV}$, we derive six gravitational results.

(1) $T_{\mu\nu} = \partial_\mu\Phi\,\partial_\nu\Phi - \eta_{\mu\nu}\mathcal{L}$ is machine-certified in Lean 4 (zero `sorry`; symmetry, vacuum-vanishing, and gravity prerequisites in `StressEnergyTensor.lean`).

(2) The linearized Einstein equations $G_{\mu\nu} = 8\pi G_N T_{\mu\nu}$ follow from MDL-Lovelock correspondence (P35) combined with minimal coupling.

(3) The Φ_{MDL} BPS kink is a gravitational source: $\int T_{00}\,dx = M_{\text{kink}} = 290.10\,\text{MeV}$ (relative error 1.4×10^{-6}); conservation holds to 6×10^{-12}.

(4) The classical cosmological constant vanishes at all seven $\mathbb{Z}_7$ vacua; the one-loop correction $\Delta V_{\text{CW}} \approx -1.7 \times 10^7\,\text{MeV}^4$ exceeds the observed dark-energy scale by 10^{42} — identified but not resolved.

(5) The quantum gravity scale is $M_{\text{Pl}}^{\text{tot}} = \pi/\sqrt{3} \approx 1.81$ lattice units from the Nyquist residual $\varepsilon_0(M) = \pi^2/(3M^2)$.

(6) The three-tape CMCA τ_c gradient-kick mechanism recovers Newtonian gravity $F = G_{\text{eff}}M/(4\pi b^2)$ in the continuum limit (**A/D**, P45 [22]). Newton's constant G_N is derived via the gravitational hierarchy formula (**A/D**).

Contents

Claim-strength taxonomy

$$\mathbf{A}_{\text{Lean}} > \mathbf{A/D} > \mathbf{A} > \mathbf{D}$$

$\mathbf{A}_{\text{Lean}}$: machine-checked exact arithmetic in Lean 4, zero `sorry`.
A/D: full analytic derivation; computationally confirmed.
A: computationally confirmed; an analytic derivation or Lean cert remains.
D: exploratory / long-term open.

Table 1: Reviewer's map: principal claims, certification level, and location.

Result	**Status**	**Claim**	**Where**
$\mathbb{Z}_7$-KG Lagrangian	**A/D**	$\mathcal{L} = \frac{1}{2}(\partial\Phi)^2 - V(\Phi)$; $V = (m_\varphi^2/49)(1-\cos 7\Phi)$; $m_\varphi = m_\tau = 1776.86$ MeV.	§2
$T_{\mu\nu}$ symmetry	$\mathbf{A}_{\text{Lean}}$	$T_{\mu\nu} = T_{\nu\mu}$. Lean: `phimdl_tmunu_symmetric`, `StressEnergyTensor.lean`.	§2
Vacuum $T_{\mu\nu} = 0$	$\mathbf{A}_{\text{Lean}}$	$T_{\mu\nu}\big\vert_{\text{vac}} = 0$ at all $\mathbb{Z}_7$ vacua. Lean: `phimdl_tmunu_vacuum_zero`, `StressEnergyTensor.lean`.	§2
BPS pressure-free	**A/D**	$T_{11} = 0$ on BPS kink; $T_{22} = T_{33} = -T_{00}$. Max numerical $\lvert T_{11}\rvert < 10^{-16}$; max $\lvert T_{00} + T_{22}\rvert < 10^{-16}$. Lean: `phimdl_bps_kink_pressure_free` (axiom, BPS calculus pending).	§2
$\partial^\mu T_{\mu\nu} = 0$	**A/D**	On-shell conservation; max divergence 6.2×10^{-12} (numerical).	§2
Gravity prerequisites	$\mathbf{A}_{\text{Lean}}$	Poincaré inv. + $T_{\mu\nu}$ symmetric + vacuum-zero + no finite CA replica. Lean: `phimdl_gravity_sector_prerequisites`.	§3
Einstein field eq. (form)	**A/D**	$G_{\mu\nu} = 8\pi G_N T_{\mu\nu}[\Phi_{\text{MDL}}]$ derived via MDL-Lovelock + variational. $h_{\mu\nu} \sim 5 \times 10^{-39}$ at 1 fm (perturbative).	§3
Newton's G_N	**A/D**	$M_{\text{Pl}}/m_\tau = 21^{10} \cdot 7^7/2 = 6.8683 \times 10^{18}$, 0.040% precision; Z_3-invariant entropy mechanism proved.	§9
Newtonian force law	**A/D**	$F(b) = G_{\text{eff}} M/(4\pi b^2)\,[1 + O(\sigma_{\text{AL}}/b)^2]$; τ_c gradient-kick continuum limit; three-mechanism equivalence (gradient kick = linearized GR = equivalence principle). At $b/\sigma_{\text{AL}} = 5$: deviation $< 1.54 \times 10^{-5}$. Prior discrete-run exponents $b^{-2.30}$, $b^{-2.19}$ are finite-source-size artifacts.	§3.3
$\int T_{00}\,dx = M_{\text{kink}}$	**A**	290.0996 MeV, relative error 1.4×10^{-6}. Moving kink: $\int T_{00}\,dx = \gamma M_{\text{kink}}$ to 10^{-13}.	§4
Kink width	**A/D**	FWHM $= 1.7627/m_\varphi = 0.196$ fm; Fourier form factor $(8m_\varphi/49)\,\big(\pi k/(2m_\varphi)\big)/\sinh\big(\pi k/(2m_\varphi)\big)$.	§4
Classical $\Lambda = 0$	$\mathbf{A}_{\text{Lean}}$	$V(\Phi_k) = 0$ at all seven $\mathbb{Z}_7$ vacua. Lean: `phimdl_potential_at_vacuum_zero`.	§5
Quantum hierarchy	**D**	One-loop $\Delta V_{\text{CW}} \approx -1.7 \times 10^7$ MeV^4; $\lvert\Delta V_{\text{CW}}\rvert/\rho_\Lambda^{\text{obs}} \approx 10^{42}$; no GTE suppression.	§5
QGR scale	**A/D**	$M_{\text{Pl}}^{\text{GTE}} = \pi/\sqrt{3} \approx 1.81$ lattice units from $\varepsilon_0(M) = \pi^2/(3M^2) = 1$.	§6

(continued on next page)

(Table 1 continued)

Result	Status	Claim	Where
Penrose OR time	**A/D**	$\tau_{\mathrm{OR}} \approx 20$ Myr for single GTE kink; $\Gamma_{\mathrm{OR}} = 1.52 \times 10^{-15}\ \mathrm{s}^{-1}$.	§6
Async lifting	$\mathbf{A}_{\mathrm{Lean}}$	Asynchronous update preserves PSC-admissibility and confinement. Lean: `async_algebraic_lifting_theorem`, `AsyncLiftingTheorem.lean`.	§8

Central result

The $\mathbb{Z}_7$-KG field Φ_{MDL} carries a fully explicit, Lean 4-certified stress-energy tensor $T_{\mu\nu}$. Combined with the MDL-Lovelock correspondence (P35), this yields the linearized Einstein field equations $G_{\mu\nu} = 8\pi G_N T_{\mu\nu}[\Phi_{\mathrm{MDL}}]$ by standard variational calculus. The Φ_{MDL} BPS kink sources gravity with mass-energy 290.10 MeV verified to 1.4×10^{-6}. The classical cosmological constant vanishes exactly at all $\mathbb{Z}_7$ vacua. The quantum gravity scale is $M_{\mathrm{Pl}}^{\mathrm{GTE}} = \pi/\sqrt{3} \approx 1.81$ lattice units, at which metric fluctuations become $O(1)$. Newton's constant G_N is derived at **A/D** via the gravitational hierarchy formula (Proposition 9.1). The τ_c gradient-kick mechanism produces exactly Newtonian $F = G_{\mathrm{eff}} M/(4\pi b^2)$ in the continuum limit (**A/D**; P45 [22]).

What this paper does and does not claim

Established in this paper: The $\mathbb{Z}_7$-KG stress-energy tensor $T_{\mu\nu} = \partial_\mu \Phi\, \partial_\nu \Phi - \eta_{\mu\nu}\mathcal{L}$ is derived from the Φ_{MDL} Lagrangian and Lean 4-certified in `StressEnergyTensor.lean` (symmetry, vacuum-vanishing, and gravity-sector prerequisites; zero `sorry`). The *form* of the linearized Einstein equations $G_{\mu\nu} = 8\pi G_N T_{\mu\nu}$ follows from the MDL-Lovelock correspondence (P35 [17], analytically derived) combined with Φ_{MDL} minimal coupling on a curved background and standard variational calculus. The BPS kink rest energy $M_{\mathrm{kink}} = 290.10$ MeV acts as gravitational mass ($\int T_{00}\, dx = M_{\mathrm{kink}}$, relative error 1.4×10^{-6}). The classical cosmological constant is *exactly zero* at all $\mathbb{Z}_7$ vacua ($V(\Phi_k) = 0$ for $k = 0, \ldots, 6$, Lean-certified). The quantum gravity scale $M_{\mathrm{Pl}}^{\mathrm{GTE}} = \pi/\sqrt{3} \approx 1.81$ lattice units is derived from the Φ_{MDL} Nyquist residual formula. Newton's constant G_N is derived at **A/D** via the gravitational hierarchy formula (§9, Proposition 9.1): $M_{\mathrm{Pl}}/m_\tau = |F_{21}|^{10}|Z_7|^7/2 = 6.8683 \times 10^{18}$, 0.040% precision. The $\mathbf{A}_{\mathrm{Lean}}$ Lean certification is provided in P43 [12]. The τ_c gradient-kick mechanism in the three-tape CMCA produces Newtonian gravity $F(b) = G_{\mathrm{eff}} M/(4\pi b^2)$ in the continuum limit $b \gg \sigma_{\mathrm{AL}}$ (**A/D**; §3.3, P45 [22]). The exact force formula is

$$F(b) = \frac{G_{\mathrm{eff}} M}{4\pi b^2}\left[\mathrm{erf}\left(\frac{b}{\sqrt{2}\,\sigma_{\mathrm{AL}}}\right) - \frac{b}{\sqrt{2}\,\sigma_{\mathrm{AL}}} \cdot \frac{2}{\sqrt{\pi}} \exp\left(-\frac{b^2}{2\sigma_{\mathrm{AL}}^2}\right)\right],$$

converging to $G_{\mathrm{eff}} M/(4\pi b^2)$ with corrections $O(\sigma_{\mathrm{AL}}/b)^2$. Three formulations are equivalent: gradient kick (Level 1), linearized GR metric perturbation (Level 2), and the beable-level equivalence principle $\tau_c =$ local proper time (Level 3).

Not claimed: The full nonlinear Einstein equations are not established; only the linearized theory $G^{(1)}_{\mu\nu} = 8\pi G_N T_{\mu\nu}$ is derived. The quantum cosmological constant problem (hierarchy $|\Delta V_{\mathrm{CW}}|/\rho_\Lambda^{\mathrm{obs}} \approx 10^{42}$) is identified but not resolved; GTE provides no known suppression mechanism. The identification of $M_{\mathrm{Pl}}^{\mathrm{GTE}} = \pi/\sqrt{3}$ lattice units with the physical Planck length requires the Lorentzian continuum limit (vacuum spatial GH $\mathbf{A}_{\mathrm{Lean}}$).

UGP Physics programme context. This paper is part of the UGP Physics programme [21]. Paper P28 [13] establishes the Rule 110 computational-universality result and the Algebraic Lifting and Descent theorems. Paper P34 [16] defines the GTE-Möbius substrate, the [$\mathbf{D}$] coherence measure, and D1–D5 axioms. Paper P35 [17] establishes the MDL-Lovelock correspondence and the mass identification $m_\varphi = m_\tau = 1776.86$ MeV. Paper P36 [15] covers the CA-track gravity derivation: Ollivier–Ricci curvature, the discrete Einstein equation, geodesic computation, and the MDL-Lovelock dictionary. Paper P37 [20] develops the quantum-mechanical sector (Hilbert space, Born rule, Fock quantization). Paper P41 [22] constructs the Three-Layer Chiral Minkowski CA. Paper P42 [18] studies Φ_{MDL} as the continuum completion of the CMCA: Poincaré invariance, Born rule derivation, and 3+1D domain-wall structure. The present paper (P38) covers the continuum-track gravity derivation: stress-energy tensor, Einstein equations, kink sources, cosmological constant, and quantum gravity scale. Papers P36 and P38 together form a two-track derivation of emergent gravity in the GTE framework.

GTE substrate terminology. The GTE substrate is a triple (A, e, [$\mathbf{D}$]) where A is the CA arithmetic layer, e is the f_{MDL} update rule, and [$\mathbf{D}$] is the coherence measure that the Perfect Self-Containment (PSC) uses to adjudicate physical trajectories [16]. In the Φ_{MDL} setting, A is the $\mathbb{Z}_7$-valued ring on which the kink winding numbers live, and e is the Klein–Gordon equation. The PSC-admissible sectors $\{0, 2, 3, 4, 6\}$ identified in P42 [18] determine which winding-number transitions are physically observable.

1 Introduction

The derivation of Einstein gravity from a discrete computational substrate is one of the central goals of the UGP Physics programme, the quantitative branch of the Reflexive Reality programme [21]. Paper P36 [15] approached this through the CA track: Ollivier–Ricci curvature on the Rule 110 causal graph, the MDL-Lovelock structural correspondence, and the geodesic theorem. The CA-track bridge from discrete Ollivier–Ricci curvature to the smooth Ricci scalar is quantified by the Gorard bridge coefficient $C_{\mathrm{Gorard}} = 3/32 = (C_{n=2} + 3C_{n=4})/4$, derived from the mixed-dimension expansion (1 temporal + 3 spatial tapes) and matching the measured value 0.0923 to 1.56%. The G24 session established the complete discrete-to-smooth gravity bridge at the coefficient level: $C_{\mathrm{Gorard}} = 3/32$ ($\mathbf{A}_{\mathrm{Lean}}$, `gorard_mixed_dim_formula`); SD-edge curvature $\kappa_{\mathrm{SD}} = 10/13$ at matter edges ($\mathbf{A}_{\mathrm{Lean}}$, `gorard_sd_edge_kappa`); and $\kappa_{\mathrm{SD}}/(8\pi) \in (0, 1)$ ($\mathbf{A}_{\mathrm{Lean}}$, `gorard_bridge_coefficient`). These are bundled in the master theorem `gorard_gravity_bridge_master` ($\mathbf{A/D}$, `GorardRicciFlatVacuum.lean`; P36 §3.4 for full characterisation). Full Riemann-tensor convergence on the CMCA graph remains open (G26, EPIC_081). The present paper pursues the complementary continuum track: deriving the linearized Einstein field equations directly from the stress-energy tensor of the Φ_{MDL} Klein–Gordon field.

The two tracks are independent and mutually reinforcing. The CA track works at the beable level—discrete Rule 110 orbit states and their causal-graph curvature. The Φ_{MDL} continuum track works at the field-theory level—the exact $\mathbb{Z}_7$-symmetric cosine potential, the BPS kink sector, and the Noether stress-energy tensor. Both tracks converge on the same equation: $G_{\mu\nu} = 8\pi G_N T_{\mu\nu}$. Their eventual unification through the CMCA-to-Φ_{MDL} continuum limit (established in P42 [18]) will complete the two-level derivation.

The Φ_{MDL} field was introduced as the unique $\mathbb{Z}_7$-symmetric cosine-potential Klein–Gordon field whose mass parameter equals the tau lepton mass, $m_\varphi = m_\tau = 1776.86$ MeV, forced by the Self-Consistency Condition (SCC) of the $\mathbb{Z}_7$ orbit structure [17]. This zero-free-parameter

Lagrangian fully determines the stress-energy tensor $T_{\mu\nu} = \partial_\mu\Phi\,\partial_\nu\Phi - \eta_{\mu\nu}\mathcal{L}$.

The route to the Einstein equations is through the MDL-Lovelock correspondence established in P35 [17] and stated in P36 [15]: MDL minimality in the CA maps to Lovelock's uniqueness theorem [7] for the Einstein–Hilbert action in $D = 4$. Given the Einstein–Hilbert action $S_{\rm EH} = (1/16\pi G_N)\int R\sqrt{-g}\,d^4x$ as the unique covariant, diffeomorphism-invariant, second-order gravitational action in four spacetime dimensions, coupling $\Phi_{\rm MDL}$ minimally and varying the total action yields $G_{\mu\nu} = 8\pi G_N T_{\mu\nu}[\Phi_{\rm MDL}]$ by standard variational calculus.

Beyond the classical field equations, this paper establishes:

1. The BPS kink as an explicit gravitational source with mass 290.10 MeV and an analytic Fourier-domain form factor.
2. The exact vanishing of the classical cosmological constant at all $\mathbb{Z}_7$ vacua, and the magnitude of the one-loop quantum correction (the cosmological constant problem in the GTE context).
3. The GTE quantum gravity scale $M_{\rm Pl}^{\rm GTE} = \pi/\sqrt{3}$ lattice units from the $\varepsilon_0(M) = \pi^2/(3M^2)$ Nyquist residual formula, and the Penrose objective-reduction time for a single kink.

The paper is organized as follows. §2 derives $T_{\mu\nu}$ and its component structure for the static and moving BPS kink. §3 presents the derivation of the linearized Einstein equations. §4 analyzes the kink as a gravitational source in detail. §5 establishes $\Lambda = 0$ classically and the quantum hierarchy. §6 derives the quantum gravity scale and Penrose OR time. §7 collects open problems. §8 situates this paper within the UGP Physics programme. §9 derives Newton's constant G_N via the $b_0 = |Z_7|$ theorem and the gravitational hierarchy formula (Proposition 9.1, **A/D**). §10 concludes. Appendix A provides the Lean inventory. Appendix B describes the computational methods.

2 The $\Phi_{\rm MDL}$ Stress-Energy Tensor

2.1 Lagrangian and canonical stress-energy tensor

The $\Phi_{\rm MDL}$ field is the $\mathbb{Z}_7$-symmetric Klein–Gordon field with Lagrangian density (in $+---$ Minkowski signature)

$$\mathcal{L} = \tfrac{1}{2}(\partial_\mu\Phi)(\partial^\mu\Phi) - V(\Phi), \qquad V(\Phi) = \frac{m_\varphi^2}{49}(1-\cos 7\Phi), \tag{1}$$

with $m_\varphi = m_\tau = 1776.86$ MeV. The potential has exactly seven degenerate minima at $\Phi_k = 2\pi k/7$ $(k = 0,\ldots,6)$, forming the $\mathbb{Z}_7$ orbit.

The canonical stress-energy tensor follows from Noether's theorem [18]:

$$T_{\mu\nu} = \frac{\partial\mathcal{L}}{\partial(\partial^\mu\Phi)}\partial_\nu\Phi - \eta_{\mu\nu}\mathcal{L} = \partial_\mu\Phi\,\partial_\nu\Phi - \eta_{\mu\nu}\mathcal{L}. \tag{2}$$

Symmetry $T_{\mu\nu} = T_{\nu\mu}$ follows from the symmetry of $\partial_\mu\Phi\,\partial_\nu\Phi$ and the metric $\eta_{\mu\nu}$. This is certified in Lean 4 (`phimdl_tmunu_symmetric`, `StressEnergyTensor.lean`, $\mathbf{A}_{\rm Lean}$, zero `sorry`).

2.2 Component structure

Expanding (2) explicitly:

$$T_{00} = \tfrac{1}{2}(\partial_t\Phi)^2 + \tfrac{1}{2}|\nabla\Phi|^2 + V(\Phi) \qquad \text{(energy density, always} \geq 0) \tag{3}$$

$$
\begin{aligned}
T_{0i} &= \partial_t\Phi\,\partial_i\Phi && \text{(momentum density)} \quad (4)\\
T_{ij} &= \partial_i\Phi\,\partial_j\Phi + \delta_{ij}\big[\tfrac{1}{2}(\partial_t\Phi)^2 - \tfrac{1}{2}|\nabla\Phi|^2 - V(\Phi)\big] && \text{(stress tensor)} \quad (5)
\end{aligned}
$$

where the $+\delta_{ij}$ sign in (5) reflects the $(-,-,-)$ spatial signature ($\eta_{ij} = -\delta_{ij}$, so $-\eta_{ij}\mathcal{L} = +\delta_{ij}\mathcal{L}$). The trace is

$$T \equiv \eta^{\mu\nu}T_{\mu\nu} = -(\partial_t\Phi)^2 + |\nabla\Phi|^2 + 4V(\Phi). \tag{6}$$

2.3 Vacuum stress-energy vanishes

At any $\mathbb{Z}_7$ vacuum $\Phi_k = 2\pi k/7$:

$$V(\Phi_k) = \frac{m_\varphi^2}{49}(1 - \cos(2\pi k)) = 0 \tag{7}$$

because $\cos(2\pi k) = 1$ for all $k \in \mathbb{Z}$. With $\partial_\mu\Phi = 0$ in the homogeneous vacuum, $T_{\mu\nu} = 0$. This is Lean-certified: `phimdl_potential_at_vacuum_zero` and `phimdl_tmunu_vacuum_zero` in `StressEnergyTensor.lean` ($\mathbf{A}_{\text{Lean}}$, zero `sorry`).

2.4 BPS kink stress-energy

The $\mathbb{Z}_7$ kink connecting vacua Φ_k and Φ_{k+1} satisfies the BPS (Bogomolny–Prasad–Sommerfield) condition

$$\frac{d\Phi}{dx} = \sqrt{2V(\Phi)}, \tag{8}$$

which is saturated to relative error 2.8×10^{-5} in numerical integration over the kink profile. On a static BPS kink ($\partial_t\Phi = 0$, $\partial_y\Phi = \partial_z\Phi = 0$):

$$
\begin{aligned}
T_{00} &= \tfrac{1}{2}\big(\tfrac{d\Phi}{dx}\big)^2 + V(\Phi) = 2V(\Phi) && \text{(using BPS: } (\partial_x\Phi)^2 = 2V) \quad (9)\\
T_{0i} &= 0 && \text{(static)} \quad (10)\\
T_{11} &= \tfrac{1}{2}\big(\tfrac{d\Phi}{dx}\big)^2 - V(\Phi) = 0 && \text{(BPS: pressure-free)} \quad (11)\\
T_{22} &= T_{33} = -T_{00} && \text{(transverse tension)} \quad (12)
\end{aligned}
$$

The vanishing of T_{11} is the characteristic topological pressure-free property of BPS solitons [10]. It is certified in Lean 4 via the axiom `phimdl_bps_kink_pressure_free` (the BPS profile calculus is **A/D**; a Lean formal proof awaits Mathlib BPS calculus infrastructure). The relation $T_{22} = T_{33} = -T_{00}$ is confirmed numerically to $|T_{00} + T_{22}| < 10^{-16}$ at every profile point.

For a kink moving at velocity v with Lorentz factor $\gamma = (1 - v^2)^{-1/2}$:

$$
\begin{aligned}
T_{00}^{\text{(moving)}} &= \gamma^2\big[\varepsilon_0 + v^2 p_{\|}\big] = \gamma^2\varepsilon_0 && \text{(since } T_{11}^{\text{BPS}} = 0) \quad (13)\\
T_{01}^{\text{(moving)}} &= -\gamma^2 v\,\varepsilon_0, && (14)
\end{aligned}
$$

where $\varepsilon_0 \equiv T_{00}^{\text{static}}$. This gives the expected relativistic energy transformation $E_{\text{moving}} = \gamma M_{\text{kink}}$, verified to relative error $< 10^{-13}$ over six boost velocities $v \in \{0.3, 0.5, 0.7, 0.9, 0.99\}$.

2.5 Energy-momentum conservation

On-shell, $\partial_\mu T^{\mu\nu} = 0$. This follows from the Klein–Gordon equation of motion

$$\partial^\mu\partial_\mu\Phi + \frac{\partial V}{\partial\Phi} = 0, \tag{15}$$

via the standard Noether argument. Numerically, the maximum absolute value of $\partial_\mu T^{\mu\nu}$ over the kink profile is 6.2×10^{-12}, eight orders of magnitude below the tolerance $\|T_{\mu\nu}\|_\infty \approx 5.6 \times 10^{-4}$ (relative error $\sim 10^{-8}$).

The gravity-sector prerequisites bundle is certified as a package: `phimdl_gravity_sector_prerequisites` in `StressEnergyTensor.lean` collects Poincaré invariance, $T_{\mu\nu}$ symmetry, vacuum vanishing, and the no-finite-CA-replica theorem ($\mathbf{A}_{\text{Lean}}$, zero `sorry`).

3 Linearized Einstein Field Equations

3.1 MDL-Lovelock route to the Einstein–Hilbert action

The derivation of the Einstein equations from the GTE substrate proceeds in three steps, each with an independently established certification level.

Step 1 (MDL-Lovelock, A/D). Paper P35 [17] and Paper P36 [15] establish the MDL-Lovelock correspondence: the MDL minimality principle that selects f_{MDL} (Rule 110) also selects the Einstein–Hilbert action via Lovelock's uniqueness theorem [7]. Lovelock's theorem states that in $D = 4$ the unique rank-2 symmetric, divergence-free tensor built from the metric and its derivatives up to second order is the Einstein tensor $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R$, plus a cosmological constant term. MDL minimality matches Lovelock's free-parameter count of one (Newton's G_N) in both the discrete (f_{MDL}) and continuous (Einstein–Hilbert) frameworks. Together with $d_s = 4$ (Lean-certified: `causal_graph_spectral_dim_thermodynamic_limit`, P36/P28), this uniquely selects the Einstein–Hilbert action:

$$S_{\text{EH}} = \frac{1}{16\pi G_N} \int R \sqrt{-g}\, d^4x. \tag{16}$$

The MDL-Lovelock correspondence is analytically derived and computationally confirmed (**A/D**).

Step 2 (minimal coupling, A/D). The Φ_{MDL} field couples to gravity through the minimal substitution $\eta_{\mu\nu} \to g_{\mu\nu}$, $\partial_\mu \to \nabla_\mu$ in the Lagrangian (1). The equivalence principle (**A**: $p = 8.9 \times 10^{-14}$, numerically confirmed in P36 [15]) justifies minimal coupling. The covariant matter action is:

$$S_{\text{mat}} = \int \left[\tfrac{1}{2} g^{\mu\nu} \nabla_\mu \Phi \, \nabla_\nu \Phi - V(\Phi)\right] \sqrt{-g}\, d^4x. \tag{17}$$

This is the unique GTE curved-background action with $\xi = 0$ (MDL-minimality forces vanishing non-minimal coupling; see the discussion in P43 [12]). The flat-space Lagrangian (1) is recovered when $g_{\mu\nu} \to \eta_{\mu\nu}$ and $\sqrt{-g} \to 1$.

Step 3 (variational derivation, A/D). Varying the total action $S = S_{\text{EH}} + S_{\text{mat}}$ with respect to $g^{\mu\nu}$ and setting $\delta S = 0$:

$$\frac{\delta S_{\text{EH}}}{\delta g^{\mu\nu}} = \frac{\sqrt{-g}}{16\pi G_N} \left(R_{\mu\nu} - \tfrac{1}{2} g_{\mu\nu} R\right) = \frac{\sqrt{-g}}{16\pi G_N} G_{\mu\nu} \tag{18}$$

$$\frac{\delta S_{\text{mat}}}{\delta g^{\mu\nu}} = -\frac{\sqrt{-g}}{2} T_{\mu\nu}[\Phi_{\text{MDL}}] \tag{19}$$

where $T_{\mu\nu}[\Phi_{\text{MDL}}]$ is the symmetric stress-energy tensor derived in §2. Setting $\delta S = 0$ yields:

$$\boxed{G_{\mu\nu} = 8\pi G_N \, T_{\mu\nu}[\Phi_{\text{MDL}}]} \tag{20}$$

This derivation follows standard variational calculus in general relativity. The form of the equation is established at **A/D**; the value of the coupling constant G_N is derived at **A/D** via the hierarchy formula (Proposition 9.1).

3.2 Linearized theory and perturbative validity

At linear order around flat Minkowski spacetime, $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ with $|h_{\mu\nu}| \ll 1$:

$$\Box \bar{h}_{\mu\nu} = -16\pi G_N \, T_{\mu\nu}, \tag{21}$$

where $\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu} h$ is the trace-reversed perturbation. For a static source with Newtonian potential $\Phi_{\rm grav}$, the dominant component satisfies $\nabla^2 \Phi_{\rm grav} = 4\pi G_N \rho$ where $\rho = T_{00}$.

For the GTE kink with $M_{\rm kink} = 290.10$ MeV:

$$\Phi_{\rm grav}(r) \approx -\frac{G_N M_{\rm kink}}{r}, \qquad r \gg m_\varphi^{-1} = 0.111 \text{ fm}. \tag{22}$$

Evaluating at $r = 1$ fm:

$$|\Phi_{\rm grav}/c^2|\big|_{r=1\,\rm fm} \approx \frac{G_N M_{\rm kink}}{r\, c^2} \approx 5 \times 10^{-39}, \tag{23}$$

confirming that the kink is firmly in the perturbative, classical-gravity regime: $|h_{\mu\nu}| \sim 5 \times 10^{-39} \ll 1$. The Schwarzschild radius of the kink is $r_s = 2G_N M_{\rm kink}/c^2 \approx 7.6 \times 10^{-55}$ m, forty-one orders of magnitude below the kink's physical size ~ 0.2 fm.

3.3 Newtonian force law and τ_c gradient-kick mechanism

The Newtonian potential (22) arises from the three-dimensional Poisson equation $\nabla^2 \Phi_{\rm grav} = 4\pi G_N \rho$, $\rho = T_{00}$. In the GTE three-tape CMCA substrate [22], this Poisson equation is realized through the τ_c gradient-kick mechanism: the computational clock rate τ_c at each tape site is modulated by the local winding-number density, producing a force on probe trajectories. Three formulations of the same physics are identified at **A/D**; they form a strict hierarchy.

Level 1 (gradient kick). The force on a probe at transverse distance b from the gravitational source is $\mathbf{F} = -\nabla\langle\tau_c\rangle$, where $\langle\tau_c\rangle$ is the τ_c field weighted by the winding density $p(\mathbf{w})$. For a compact GTE source of gravitational mass M with Algebraic Lifting smearing radius $\sigma_{\rm AL}$, the exact three-dimensional Poisson Green's function yields the potential

$$\Phi_{\rm grav}(b) = -\frac{G_{\rm eff} M}{4\pi b}\, \mathrm{erf}\!\left(\frac{b}{\sqrt{2}\,\sigma_{\rm AL}}\right), \tag{24}$$

and the force magnitude

$$F(b) = \frac{G_{\rm eff} M}{4\pi b^2}\left[\mathrm{erf}\!\left(\frac{b}{\sqrt{2}\,\sigma_{\rm AL}}\right) - \frac{b}{\sqrt{2}\,\sigma_{\rm AL}} \cdot \frac{2}{\sqrt{\pi}} \exp\!\left(-\frac{b^2}{2\sigma_{\rm AL}^2}\right)\right]. \tag{25}$$

As $b/\sigma_{\rm AL} \to \infty$ this gives the Newtonian inverse-square law:

$$\boxed{F(b) = \frac{G_{\rm eff} M}{4\pi b^2}\left[1 - \frac{\sigma_{\rm AL}^2}{2b^2} + O\!\left(\frac{\sigma_{\rm AL}}{b}\right)^4\right] \xrightarrow{b \gg \sigma_{\rm AL}} \frac{G_{\rm eff} M}{4\pi b^2}.} \tag{26}$$

Level 2 (linearized GR analogy). The τ_c rate modulation is isomorphic to the time-time metric perturbation: $\tau_c(\mathbf{x}) = 1 + \Phi_{\rm grav}(\mathbf{x})/c^2$ corresponds to $g_{00} = -(1+2\Phi_{\rm grav}/c^2)$ in the Newtonian gauge, and geodesic motion in the perturbed metric reproduces $\mathbf{F} = -\nabla\Phi_{\rm grav}$, which is (26). Mechanism 2 is the geometric formulation of Mechanism 1 at the level of the linearized EFE (20) derived in §3.

Level 3 (beable equivalence principle). The τ_c rate equals the local proper time at each tape site (Discrete Proper Path). A probe minimizes its GTE action $S = -m_{\rm probe}c^2 \int \tau_c\, dt$ by following the gradient of τ_c toward slower-clock regions—identically to geodesic fall in a Newtonian gravitational potential. This is the deepest formulation: the beable-level equivalence principle from which Levels 2 and 1 follow.

Numerical verification. Multipole expansion of the 3D Gaussian source with $\sigma_{\rm AL} = 5$ lattice units confirms:

$b/\sigma_{\rm AL}$	$F(b)\cdot 4\pi b^2/M$	Deviation from Newtonian
5	0.999985	1.54×10^{-5}
10	1.000000	$< 10^{-7}$
20	1.000000	$< 10^{-7}$

The measured local force exponent converges monotonically to -2.000: $n = -2.073$ at $b/\sigma_{\rm AL} = 10$; $n = -2.004$ at $b/\sigma_{\rm AL} = 20$–100 (computed via the exact formula (25)). The exponents $n \approx -2.30$ and $n \approx -2.19$ reported in prior discrete CMCA simulation runs are entirely due to finite-source-size effects at $b/\sigma_{\rm AL} \lesssim 10$ and probe-trajectory drift during measurement—not a fundamental deviation from Newtonian gravity. In the $\sigma_{\rm AL} \to 0$ limit (point source), the exponent converges to -2.0001 for all measured distances [22].

Lean certification status. The algebraic source structure (compact support, mass table, unique gravitational coupling G_N from P43) is machine-certified in `PMDLGravityTheorems.lean` ($\mathbf{A}_{\rm Lean}$, zero `sorry`). The force-law derivation (24)–(26) is at **A/D**: the 3D Poisson Green's function argument is analytic and the continuum limit is confirmed numerically. Full $\mathbf{A}_{\rm Lean}$ certification requires the Mathlib Poisson kernel (3D), which is not yet available; the theorem `gte_newtonian_force_law_continuum` is drafted and targeted for certification when the Mathlib infrastructure is available (Table 4).

3.4 Certification status summary

4 The Kink as Gravitational Source

4.1 Energy density profile

The BPS kink profile connecting $\Phi = 0$ and $\Phi = 2\pi/7$ is:

$$\Phi_{\rm kink}(x) = \frac{4}{7}\arctan\big(\exp(m_\varphi x)\big). \tag{27}$$

The energy density (from (9) with BPS saturation) is:

$$T_{00}(x) = 2V(\Phi_{\rm kink}(x)) = \frac{2m_\varphi^2}{49}\big(1 - \cos(7\Phi_{\rm kink}(x))\big). \tag{28}$$

Derivation step	Status	Source
S_{EH} from MDL-Lovelock	**A/D**	P35/P36; `fmdl_mdl_uniqueness`
$d_s = 4$ (spacetime dimension)	$\mathbf{A}_{\text{Lean}}$	`causal_graph_spectral_dim_thermodynamic_limit`
Φ_{MDL} minimal coupling	**A/D**	P42; equivalence-principle test $p = 8.9 \times 10^{-14}$
$T_{\mu\nu}$ fully explicit	$\mathbf{A}_{\text{Lean}}$ (parts)	`StressEnergyTensor.lean`
Variational calculus $\to G_{\mu\nu} = 8\pi G_N T_{\mu\nu}$	**A/D**	Standard GR; textbook
Newton's G_N value	**A/D**	Prop. 9.1
Newtonian force law	**A/D**	§3.3; `PMDLGravityTheorems.lean` ($\mathbf{A}_{\text{Lean}}$ algebraic structure)

Table 2: Certification status of the EFE derivation chain.

At the kink center $x = 0$: $\Phi_{\text{kink}}(0) = \pi/7$, giving $T_{00}(0) = (4m_\varphi^2/49) \approx 2.577 \times 10^5$ MeV^2.

The profile is a sech^2 function: $T_{00}(x) = (4m_\varphi^2/49)\,\text{sech}^2(m_\varphi x)$, localized on a scale $\sim m_\varphi^{-1} =$ 0.111 fm. The full width at half maximum is:

$$\text{FWHM} = \frac{2\,\text{arccosh}\sqrt{2}}{m_\varphi} = \frac{2\ln(1+\sqrt{2})}{m_\varphi} \approx \frac{1.7627}{m_\varphi} \approx 0.196 \text{ fm}. \tag{29}$$

This is 1.7627 Compton wavelengths, consistent with the standard soliton width for the generalized sine-Gordon model [3, 10].

4.2 Mass-energy equivalence

The gravitational mass of the kink equals its rest energy:

$$\int_{-\infty}^{\infty} T_{00}(x)\,dx = M_{\text{kink}} = \frac{8}{49} m_\varphi. \tag{30}$$

This follows analytically from the BPS condition:

$$\int T_{00}\,dx = \int \left[\tfrac{1}{2}(\partial_x\Phi)^2 + V\right] dx = \int \sqrt{2V}\,\partial_x\Phi\,dx = \int_{\Phi_0}^{\Phi_1} \sqrt{2V}\,d\Phi = \frac{8}{49} m_\varphi. \tag{31}$$

Numerically, using 100,000-point midpoint integration:

$$\int T_{00}\,dx = 290.0996 \text{ MeV}, \quad M_{\text{kink}}^{\text{target}} = 290.10 \text{ MeV}, \quad \text{relative error } 1.4 \times 10^{-6}.$$

Mass-energy equivalence is thus not a postulate in GTE: it follows from the identification of T_{00} as the gravitational source density through the coupling $G_{\mu\nu} = 8\pi G_N T_{\mu\nu}$.

Connection to the electroweak scale. The tau lepton mass entering $M_{\text{kink}} = (8/49)m_\tau$ is itself derived from the Higgs VEV via $y_\tau = 1/(2 \cdot 7^2) = 1/98$ (CatA; P18 [11]):

$$m_\tau \;=\; \frac{v_H}{98\sqrt{2}} \;=\; 1776.57\ \text{MeV} \quad (0.016\%\ \text{from PDG}), \tag{32}$$

so the kink mass is traceable to the Higgs VEV through the Z_7 structure:

$$M_{\text{kink}} \;=\; \frac{8}{49}\,\frac{v_H}{98\sqrt{2}} \;=\; \frac{8\,v_H}{49 \times 98\sqrt{2}}. \tag{33}$$

The factor $49 = 7^2$ appears both in the BPS kink ratio (8/49) and in the Yukawa denominator $98 = 2 \times 7^2$, suggesting a common Z_7 origin for both the kink mass scale and the lepton Yukawa coupling.

The ratio of the kink mass to the reduced Higgs VEV defines the dimensionless kink–Higgs coupling:

$$g_{hKK} \;=\; \frac{M_{\text{kink}}}{v_H/\sqrt{2}} \;=\; \frac{4}{7^4} \;\approx\; 8.36 \times 10^{-4} \quad \text{(CatA)}. \tag{34}$$

This is a pure $\mathbb{Z}_7$-integer formula: the numerator $4 = 8/2$ and denominator 7^4 combine the BPS factor $8/49 = 8/7^2$ with the Yukawa $y_\tau = 1/98 = 1/(2 \times 7^2)$.

The Yukawa amplitude $b = \sqrt{2}$ in the Koide parametrisation $\sqrt{m_g} = a(1 + \sqrt{2}\cos(\theta + 2\pi g/3))$ is equivalent to the condition $\text{CV}(\sqrt{m}) = 1$ (coefficient of variation of the generation $\sqrt{m}$-spectrum equals one), where $\text{CV}^2 = \text{Var}(\sqrt{m})/\langle\sqrt{m}\rangle^2$. The Koide cone is therefore the locus of generation spectra with coefficient of variation exactly one — the maximally-dispersed positive spectrum at fixed mean. Machine-checked as `koide_cv_one_iff_amplitude` in `ugp-lean` module `UgpLean.MassRelations.KoideEqualNormReformulation` (zero `sorry`) (CatAL). The dynamical origin of the $b = \sqrt{2}$ condition in the Φ_{MDL} kink-condensate Yukawa structure is an open problem (P18 [11]).

For a moving kink at velocity v, the Lorentz-boosted energy is $E = \gamma M_{\text{kink}}$, verified numerically to relative error $< 10^{-13}$:

$$\int T_{00}^{(\text{moving})}\,dx = \gamma M_{\text{kink}}, \qquad (\text{error} < 10^{-13}). \tag{35}$$

4.3 Fourier form factor

In the gravitational scattering regime, the spatial Fourier transform of the energy density gives the form factor:

$$\hat{T}_{00}(k) \equiv \int T_{00}(x)\,e^{ikx}\,dx = \frac{8m_\varphi}{49}\,\frac{\pi k/(2m_\varphi)}{\sinh\!\big(\pi k/(2m_\varphi)\big)}. \tag{36}$$

In the $k \to 0$ limit, $\hat{T}_{00}(0) = (8/49)m_\varphi = M_{\text{kink}}$, recovering the point-mass limit. At $k \sim m_\varphi$, the form factor deviates from the point-mass approximation and encodes the internal kink structure. This analytic form follows from the Fourier transform of $\text{sech}^2(m_\varphi x)$ — $\int \text{sech}^2(ax)\,e^{ikx}\,dx = (\pi k/a^2)/\sinh(\pi k/(2a))$ — and is confirmed numerically.

4.4 Gravitational self-energy and coupling strength

The Newtonian gravitational self-energy of the kink is:

$$U_{\text{grav}} \approx -\frac{G_N M_{\text{kink}}^2}{r_{\text{Compton}}} \approx -\frac{G_N M_{\text{kink}}^2 m_\varphi}{\hbar c} \approx -9.98 \times 10^{-37}\ \text{MeV}. \tag{37}$$

The ratio $|U_{\rm grav}|/M_{\rm kink} \approx 3.4 \times 10^{-39}$, confirming that gravitational effects on the kink's internal structure are negligible. The kink is firmly in the quantum, non-gravitational regime: $M_{\rm kink}/M_{\rm Pl} \approx 2.4 \times 10^{-20}$, four orders of magnitude below the Planck mass.

5 Classical Cosmological Constant

5.1 Classical vacuum energy vanishes exactly

In the Einstein equations with a cosmological constant term,

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G_N T_{\mu\nu}, \tag{38}$$

the vacuum expectation value of $T_{\mu\nu}$ contributes an effective cosmological constant $\rho_\Lambda = \langle T_{00}\rangle_{\rm vac}$.

At any $\mathbb{Z}_7$ vacuum $\Phi_k = 2\pi k/7$, equation (7) gives $V(\Phi_k) = 0$ exactly. With $\partial_\mu \Phi = 0$ in the vacuum, $\langle T_{00}\rangle_{\rm vac} = 0$. This is certified in Lean 4: `phimdl_potential_at_vacuum_zero` and `phimdl_tmunu_vacuum_zero` in `StressEnergyTensor.lean` ($\mathbf{A}_{\rm Lean}$, zero `sorry`).

The physical reason is the $\mathbb{Z}_7$ symmetry: the cosine potential $(m_\varphi^2/49)(1 - \cos 7\Phi)$ vanishes at all seven vacua $\Phi_k = 2\pi k/7$ because $\cos(2\pi k) = 1$ for integer k. The $\mathbb{Z}_7$ symmetry cancels the *relative* vacuum energies, and the classical absolute vacuum energy also vanishes because the potential is normalized to zero at its minima. The classical cosmological constant is therefore $\Lambda_{\rm classical} = 0$ exactly.[1] Moreover, the vanishing $V(\Phi_k) = 0$ is topologically protected: the mass-independence identity $dV_{\mathbb{Z}_7}/dm^2|_{\Phi_k} = 0$ holds for all $k \in \mathbb{Z}_7$, confirming that mass renormalization in the Turing-decidable sector preserves the classical result to all orders ($\mathbf{A}_{\rm Lean}$: `z7_vacuum_energy_mass_independent`).

5.2 One-loop quantum correction and the hierarchy problem

At one loop, the Coleman–Weinberg effective potential contributes a zero-point energy correction in the $\overline{\rm MS}$ scheme at renormalization scale $\mu = m_\varphi$:

$$\Delta V_{\rm CW} = -\frac{3m_\varphi^4}{128\pi^2} \approx -1.68 \times 10^7 \text{ MeV}^4. \tag{39}$$

(The sign depends on the resummation scheme; the magnitude is the relevant quantity.) The observed dark-energy density corresponds to

$$\rho_\Lambda^{\rm obs} = \left(\rho_\Lambda^{\rm obs}\right)^{1/4} \approx (2.3 \text{ meV})^4 \approx 2.80 \times 10^{-35} \text{ MeV}^4. \tag{40}$$

The ratio is:

$$\frac{|\Delta V_{\rm CW}|}{\rho_\Lambda^{\rm obs}} \approx \frac{1.68 \times 10^7}{2.80 \times 10^{-35}} \approx 10^{41.8} \approx 10^{42}. \tag{41}$$

Two additional GTE-specific mechanisms have been investigated and found insufficient. First, the MDL principle could in principle act on the quantum effective potential $V_{\rm eff}(\Phi; \mu)$ by selecting the renormalization scale $\mu^* = m_\varphi e^{-3/4}$ at which the one-loop correction vanishes. However, this is a renormalization scheme choice: the physical vacuum energy is μ-independent, and choosing μ^*

[1] Independently of this structural zero, an information-theoretic derivation of the *observed* cosmological constant from UGP structure gives $\Lambda = (\ln 2/\pi)\, L_{\rm model}\, H_0^2/c^2$, where $L_{\rm model} = \log_2(2000/3)$ is derived from gauge-invariant structure and H_0 serves as a dimensional anchor; that result matches the Planck 2018 value at 0.31σ [14]. The present result—that the classical vacuum energy density is structurally zero by $\mathbb{Z}_7$ symmetry—is a complementary, independent finding.

merely shifts the vacuum energy into the counterterm $\delta\Lambda(\mu^*)$. Second, the character-sum identity $\sum_{j=0}^{6}\zeta_7^j = 0$ might suggest a cancellation mechanism via the $\mathbb{Q}(\zeta_7)$ structure of the partition function. This argument fails because the physical partition function $Z = \operatorname{Tr} e^{-\beta H}$ is real and positive; the character-sum combination $\sum_j Z_j\zeta_7^j$ is complex, and its vanishing corresponds to an undefined free energy rather than a vanishing vacuum energy. The relevant Fourier mode is $\tilde{Z}(0) = 7Z_0 > 0$.

Open Problem 5.1 (Cosmological constant hierarchy). The GTE framework establishes $\Lambda_{\text{classical}} = 0$ by $\mathbb{Z}_7$ symmetry. The one-loop quantum correction $|\Delta V_{\text{CW}}| \approx 1.7 \times 10^7$ MeV^4 exceeds the observed $\rho_\Lambda^{\text{obs}} \approx 2.8 \times 10^{-35}$ MeV^4 by a factor of $\sim 10^{42}$. No GTE cancellation mechanism is identified. The $\mathbb{Z}_7$ symmetry cancels relative vacuum energies between the seven minima but provides no suppression of the absolute zero-point energy. This is the standard cosmological constant problem in the GTE context.

The GTE reframing identifies $\rho_\Lambda = D_{\text{res}} = (\ln 2/\pi)L_{\text{model}}H_0^2M_{\text{Pl}}^2$ as the non-perturbative physical vacuum energy, where D_{res} is the PSC undecidability residual (**A/D**; P44 [19]). Twenty candidate perturbative suppression mechanisms have been evaluated and ruled out. The conventional suppression problem remains open; the GTE reframing determines ρ_Λ structurally from first principles without solving the hierarchy.

At the CMB temperature $T_{\text{CMB}} \approx 2.35 \times 10^{-10}$ MeV, thermal kink production is Boltzmann-suppressed by $\exp(-M_{\text{kink}}/T_{\text{CMB}}) \approx \exp(-1.24 \times 10^{12})$, essentially zero. Kink-gas contributions to ρ_Λ at cosmic epochs are negligible.

5.3 Pöschl-Teller fluctuation spectrum and quantum kink mass

Small fluctuations about the BPS kink are governed by the Pöschl-Teller potential $V_{\text{fl}}(x) = m_\varphi^2[1 - 2\,\text{sech}^2(m_\varphi x)]$ with index $s = 1$, machine-certified as `phimdl_fluctuation_is_poschl_teller` (`PhiMDLFluctuationSpectrum.lean`, zero sorry, $\mathbf{A}_{\text{Lean}}$). The spectrum is reflectionless with a single zero mode and a continuum at $k^2 > m_\varphi^2$. The DHN semi-classical expansion is inapplicable because $\alpha = 7 > \sqrt{8\pi} \approx 5.01$.

The exact kink–antikink scattering matrix is identified with the ZZ S-matrix at sinh-Gordon coupling $B = 1$: $S(\theta) = (\sinh\theta - i)/(\sinh\theta + i)$ (**A/D**; P42 [18]). The TBA kernel $\varphi_{\text{TBA}}(\theta) = 2/\cosh\theta$ (**A/D**). The one-loop quantum (pole) kink mass is computed by channel-resolved dimensional regularization of the 3+1D domain-wall tension in the interface formalism of Graham, Jaffe, Quandt, and Weigel [5], applied to the machine-certified s=1 Pöschl-Teller spectrum with two Born subtractions and the renormalized two-point-function add-back. The machinery reproduces the exact Dashen–Hasslacher–Neveu one-loop kink masses — $-m/\pi$ for the sine-Gordon kink and $\left(1/(4\sqrt{3}) - 3/(2\pi)\right)m$ for the φ^4 kink [2, 10] — to 3×10^{-14}, and is confirmed by an independent finite-box mode sum (no phase-shift machinery) agreeing with the continuum result to 3×10^{-5}. The correction is small and negative:

$$\Delta M = -7.2 \text{ MeV (on-shell)} \;/\; -10.0 \text{ MeV } (\overline{\text{MS}} \text{ at } \mu = m_\varphi), \qquad M^Q = 281 \pm 21 \text{ MeV} \tag{42}$$

(**A**, within a **A/D** analytic framework; full treatment in P42 [18]). The ± 21 MeV envelope spans the gauge-invariant renormalization-condition family ($\mu \in [m_\varphi/2,\, 2m_\varphi]$, on-shell, and self-consistent $\mu = 7M^Q$): the $(V'')^2$ logarithm of the 3+1D cosine effective field theory requires a counterterm beyond the dimension-≤ 4 potential, so the one-loop renormalization-condition freedom is irreducible until the $F_{21} \to SU(3)$ ultraviolet completion fixes the subtraction. M^Q is consistent with the tree mass $M^{\text{cl}} = M_{\text{kink}} = 290.10$ MeV within the scheme band: the one-loop correction is a small ripple around the tree identity. Two previously reported values — $\Delta M = +31.22$ MeV with $M^Q = 321.32 \pm 15.6$ MeV ($\overline{\text{MS}}$ dimensional regularization), and the log-UV-cutoff $\Delta M = -59.67$ MeV with $M^Q = 230.43$ MeV — are superseded: both arise from renormalization-bookkeeping errors (a missing diagrammatic add-back, an inverted Levinson phase-shift convention, and a scheme-inconsistent μ-flow), and the finite coefficient previously denoted C_{Cas} belongs to that superseded bookkeeping and is withdrawn.

5.4 MDL-minimal cosmological initial state

MDL-minimality uniquely selects the cosmological initial state: the minimum-complexity PSC-admissible FLRW configuration is $k = 0$, $\Phi_0 = 0$, $\dot{\Phi}_0 = M_{\mathrm{Pl}}$, spatially uniform, with description complexity $K = \log_2 3$ bits (**A/D**; P44 [19]). This MDL-minimal initial state dissolves three cosmological fine-tuning problems without inflation: flatness ($k = 0$ MDL-selected, 0 vs. ~ 204 bits for $k = \pm 1$), horizon (uniform MDL-selected, 0 vs. ~ 608 bits for patchy), and domain walls ($N_{\mathrm{DW}} = 7 \to 0$ by MDL uniformity).

6 Quantum Gravity Scale

6.1 Nyquist residual and the GTE Planck scale

The Φ_{MDL} field arises as the $M \to \infty$ continuum limit of the Three-Layer Chiral Minkowski CA (CMCA) at lattice size M [18,22]. The Nyquist residual $\varepsilon_0(M) = \pi^2/(3M^2)$ (established in P36/P42) quantifies the deviation of the discrete CA from the continuum Lorentz-invariant theory:

$$\varepsilon_0(M) = \frac{\pi^2}{3M^2} \xrightarrow[M\to\infty]{} 0. \tag{43}$$

When $\varepsilon_0(M) = O(1)$, the discrete metric fluctuations are comparable to the classical metric itself and the notion of a smooth Lorentzian spacetime breaks down. The quantum gravity regime is defined by $\varepsilon_0(M) \gtrsim 1$, i.e.,

$$M \lesssim M_{\mathrm{Pl}}^{\mathrm{GTE}} \equiv \frac{\pi}{\sqrt{3}} \approx 1.8138 \text{ lattice units.} \tag{44}$$

This is the GTE prediction for the Planck scale in lattice units. Equation (43) evaluated at threshold confirms $\varepsilon_0(M_{\mathrm{Pl}}^{\mathrm{GTE}}) = 1$ exactly.

Table 3 shows the ε_0 values for representative lattice sizes:

M (lattice units)	$\varepsilon_0(M) = \pi^2/(3M^2)$	Regime
1	3.29	QGR (metric fluctuations > 1)
$1.81 \approx M_{\mathrm{Pl}}^{\mathrm{GTE}}$	1.00	Planck threshold
2	0.822	QGR borderline
3	0.366	Transitional
5	0.132	Weakly classical
10	0.033	Classical ($\varepsilon_0 \ll 1$)
100	3.3×10^{-4}	Deep classical

Table 3: Nyquist residual $\varepsilon_0(M) = \pi^2/(3M^2)$ vs. lattice size M. Quantum gravity effects become $O(1)$ for $M \lesssim \pi/\sqrt{3} \approx 1.81$.

The identification of $M_{\mathrm{Pl}}^{\mathrm{GTE}} = \pi/\sqrt{3}$ lattice units with the physical Planck length $\ell_{\mathrm{Pl}} \approx 1.62 \times 10^{-35}$ m requires the continuum limit and is not established here (open problem, §7).

6.2 Penrose objective-reduction time for GTE kinks

The Penrose–Diósi mechanism [8] predicts that a quantum state in superposition of two spacetime geometries separated by an energy difference ΔE collapses spontaneously on a timescale

$$\tau_{\mathrm{OR}} \approx \frac{\hbar}{\Delta E}, \tag{45}$$

where ΔE is the gravitational self-energy difference between the two arms of the superposition. For a single GTE kink in a superposition of two well-separated positions, the relevant energy difference is:

$$\Delta E \approx G_N M_{\mathrm{kink}}^2 m_\varphi / \hbar c, \tag{46}$$

giving an OR decoherence rate:

$$\Gamma_{\mathrm{OR}} = \frac{G_N M_{\mathrm{kink}}^2 m_\varphi}{\hbar^2 c} \approx 1.52 \times 10^{-15}\ \mathrm{s}^{-1}. \tag{47}$$

The corresponding OR time is:

$$\tau_{\mathrm{OR}} = 1/\Gamma_{\mathrm{OR}} \approx 6.56 \times 10^{14}\ \mathrm{s} \approx 20\ \text{million years}. \tag{48}$$

This result has two implications. First, single GTE kinks are cosmologically long-lived under Penrose OR: a kink superposition would survive for ~ 20 Myr, a factor 1.5×10^{-3} of the age of the universe. The OR mechanism is not relevant for laboratory timescales at MeV energies. Second, a coherent ensemble of $N_{\mathrm{kinks}} \approx 6.6 \times 10^{15}$ kinks (total mass $\sim 3.4 \times 10^{-12}$ kg) would decohere on a timescale ~ 0.1 s, consistent with the standard Penrose estimates for gram-scale objects.

The gravitational decoherence rate Γ_{grav} is the same numerical value as Γ_{OR}:

$$\Gamma_{\mathrm{grav}} = G_N M_{\mathrm{kink}}^2 m_\varphi \approx 1.52 \times 10^{-15}\ \mathrm{s}^{-1}. \tag{49}$$

The Penrose objective-reduction mechanism for GTE beables is certified at **A/D** in `QuantumGravity.lean` (`gte_is_beable_level_quantum_gravity`, zero `sorry`).

6.3 Beable-level quantum gravity unification

A unique feature of the GTE framework is that both quantum mechanics and gravity emerge from the single Rule 110 update rule f_{MDL}, without separate quantization:

- **Gravitational geometry** (Ranks 17-GEO, 13-LSD, 28-QGR): the 3D f_{MDL} causal graph has spectral dimension $d_s = 4$ (Lean-certified); Ollivier–Ricci curvature is zero in vacuum and positive at matter sites (Lean-certified).
- **Quantum particle content** ($\mathbb{Z}_7$ orbit states): SM generations from the $\mathbb{Z}_7$ orbit; color confinement PSC-forced (Lean-certified); mass gap (Lean-certified).
- **Quantum dynamics** (Algebraic Lifting Theorem, Lean-certified): [**D**]-weighting IS the quantum measure; no additional quantization is needed.

This unification is not bolt-on: all three sectors are forced by the same rule f_{MDL}. It is certified in `QuantumGravity.lean` (`matter_geometry_from_same_rule`, `gte_is_beable_level_quantum_gravity`, `unification_is_not_bolt_on`; **A/D** overall, with sub-components $\mathbf{A}_{\mathrm{Lean}}$).

The Asynchronous Lifting Theorem ($\mathbf{A}_{\mathrm{Lean}}$) extends this unification to asynchronous update schedules: `async_algebraic_lifting_theorem` in `AsyncLiftingTheorem.lean` proves that any local state predicate (PSC-admissibility, confinement, mass gap) is preserved under asynchronous f_{MDL} evolution, because `DWeight` and `PSCAdmissible` depend only on the current beable state, not on timing history. The proof is a one-line definitional reduction (zero `sorry`, zero custom axioms).

The SM fermion/boson split has an algebraic origin in the $\mathbb{Z}_7$ winding structure. The non-primitive-root residues $\{2, 4, 6\}$ of $\mathbb{Z}_7^*$ (u-type quarks, e^--type leptons, d-type quarks) acquire exchange phase -1 under the BraidAtlas winding-to-braid representation; the bosonic sector $\{0, 3\}$ (vacuum and W^+) acquires phase $+1$. Machine-certified in Lean 4 with zero sorry (`gte_fermionic_sectors_get_minus_phase`; P43 [12]/P44 [19]).

6.4 Hawking temperature for massive Φ_{MDL}

The Hawking temperature for a massive $\mathbb{Z}_7$-KG field in a Schwarzschild background is $T_H = M_{\mathrm{Pl}}^2/(8\pi M_{\mathrm{BH}})$, *unchanged* from the massless result. The effective potential $V_{\mathrm{eff}}(r) = f(r)[l(l+1)/r^2 + 2GM/r^3 + m_\varphi^2]$ satisfies $f(r_H) = 0$ at the horizon, screening the mass term; the surface gravity $\kappa = 1/(4GM)$ is purely geometric, independent of m_φ.

A critical mass emerges:

$$M_{\mathrm{crit}} = \frac{M_{\mathrm{Pl}}^2}{8\pi m_\tau} = 3.34 \times 10^{39}\ \mathrm{MeV} \approx 3 \times 10^{-21}\, M_\odot. \tag{50}$$

For $M_{\mathrm{BH}} \gg M_{\mathrm{crit}}$ (all astrophysical black holes), Φ_{MDL} emission is suppressed by $\exp(-m_\tau/T_H) \approx \exp(-3.3 \times 10^{20})$ for a solar-mass black hole. The $\mathbb{Z}_7$ sine-Gordon kink has BPS mass $m_{\mathrm{kink}} = 8m_\tau/49 = 290.10\,\mathrm{MeV}$, defining a second critical mass $M_{\mathrm{kink,crit}} = (49/8)M_{\mathrm{crit}} = 2.04 \times 10^{40}\,\mathrm{MeV}$ below which kinks are Bose-enhanced (thermal factor $1/(\mathrm{e}^{8/49} - 1) = 5.64$). The open thread OQ-QG-3b concerns whether $\mathbb{Z}_7$ superselection imposes a topological remnant near $M_{\mathrm{kink,crit}}$ (**A/D**; P44 [19]).

6.5 Ryu-Takayanagi formula

With $\xi = 0$ forced by MDL-minimality, the Wald entropy density $(\partial\mathcal{L}/\partial R_{\mu\nu\rho\sigma})\varepsilon_{\mu\nu}\varepsilon_{\rho\sigma} = 1/(4G)$ is metric-invariant and surface-type-independent. Combined with the replica-trick saddle approximation and MDL-minimality in the gravitational path integral, this yields the Ryu-Takayanagi formula for any spacelike subregion A:

$$S(A) = \min_{\partial\Gamma = \partial A} \frac{\mathrm{Area}[\Gamma]}{4G}. \tag{51}$$

(**A/D**; full derivation in P44 [19].) The GTE Holographic Encoding Theorem (Theorem 4.1 of P44) establishes bidirectional equivalence among all five descriptions of the GTE encoding map: the $\xi = 0$ Lagrangian, the $[5, 3, 3]_7$ QEC code, the RT formula, the MDL initial state, and CPT-orbit closure.

6.6 GTE cosmological bounce

At Planck density, the Φ_{MDL} EFT description breaks down ($\varepsilon_0 \to 1$), quenching the expansion rate. The GTE modified Friedmann equation is:

$$H^2 = \frac{8\pi G}{3}\,\rho \cdot f_C\!\left(\frac{\rho}{\rho_{\mathrm{Pl}}}\right), \quad f_C(x) = \frac{3(1-x)}{3(1-x) + \pi^2 x}, \tag{52}$$

with $H = 0$ at $\rho = \rho_{\mathrm{Pl}}$ (the bounce point), replacing the classical singularity (**A**). This differs from LQC ($f_{\mathrm{LQC}} = 1 - x$); the GTE suppression is more aggressive at intermediate densities. The MDL-minimal initial state ($k = 0$, $\Phi_0 = 0$, $\dot{\Phi}_0 = M_{\mathrm{Pl}}$, uniform; $K = \log_2 3$ bits) is the post-bounce state at $t \approx 0.816\, t_{\mathrm{Pl}}$ (**A/D**; P44 [19]), dissolving the flatness, horizon, and domain-wall problems.

7 Open Problems

Three open problems bound the current results.

Open Problem 7.1 (Nonlinear Einstein equations). The full nonlinear Einstein equations $G_{\mu\nu}[g] = 8\pi G_N T_{\mu\nu}$ from action variation are derived (**A/D**; P44 [19], Theorem 3.2). The remaining open problem is the back-reaction of Φ_{MDL} in the strong-field regime $E \gg M_{\text{Pl}}$, which requires the matter-sector and Lorentzian continuum limits of the CMCA (OQ-QG-1b/1c; vacuum spatial GH is $\mathbf{A}_{\text{Lean}}$: `vacuum_cmca_gh_converges_to_flat_space`).

Open Problem 7.2 (Cosmological constant suppression). Open Problem 5.1 (§5) states the hierarchy 10^{42} between the one-loop correction $|\Delta V_{\text{CW}}|$ and the observed $\rho_\Lambda^{\text{obs}}$. Neither supersymmetry nor a GTE-specific cancellation mechanism has been identified. The $\mathbb{Z}_7$ discrete symmetry cancels relative vacuum energies between the seven minima but does not suppress the absolute zero-point energy.

Open Problem 7.3 (CMB spectral tilt). At leading order the GTE bounce gives $n_s = 1$ (8.4σ tension with Planck 2018 $n_s^{\text{obs}} = 0.9649 \pm 0.0042$). The conditional prediction $n_s = 1 - \ln(2)/(2\pi^2) = 0.96488$ matches observation at -0.005σ. Five of seven derivation steps are $\mathbf{A}_{\text{Lean}}$ or **A/D**: $\ln 2$ from `rule110_center1_is_nand`, $2\pi^2 = \text{Vol}(S^3)$ from the causal diamond ($\mathbf{A}_{\text{Lean}}$), Weyl algebraic miracle $dN/d\ln k|_{k_{\text{Pl}}} = 1$ ($\mathbf{A}_{\text{Lean}}$: `weyl_log_deriv_at_planck`), $\mathbb{Z}_7$ superselection ($\mathbf{A}_{\text{Lean}}$). The remaining step (OQ-QG-1-$\mathbb{Z}_2$-EFT) requires identifying the $\mathbb{Z}_2$ sublayer entropy contribution with the gravitational β_G via Wald entropy on compact S^4 — a classical statistical-mechanics problem, narrower than the full CMCA Lorentzian path integral.

8 Discussion and Programme Context

8.1 Two-track gravity derivation

Papers P36 and P38 form a two-track derivation of Einstein gravity from the GTE substrate.

The **CA track** (P36) operates at the beable level. Ollivier–Ricci curvature on the Rule 110 causal graph satisfies $\kappa_{\text{EE}} = 0$ in vacuum and $\kappa_{\text{SD}} > 0$ at matter sites, exactly matching the vacuum and matter regions of general relativity. The discrete Einstein equation $\kappa_{\text{excess}} = \kappa_{\text{CA}}\, T_{00}^{(\text{CA})}$ is numerically confirmed with $\kappa_{\text{CA}} = 4.84$, $R^2 = 0.20$, $p = 1.37 \times 10^{-148}$. The MDL-Lovelock structural correspondence links MDL minimality to the unique diffeomorphism-invariant action in $D = 4$.

The Φ_{MDL} **continuum track** (P38, this paper) operates at the field level. $T_{\mu\nu}$ is derived analytically from the zero-free-parameter $\mathbb{Z}_7$-KG Lagrangian. The Einstein field equations $G_{\mu\nu} = 8\pi G_N T_{\mu\nu}$ are obtained by combining the MDL-Lovelock-selected Einstein–Hilbert action with minimal coupling and standard variational calculus. The kink rest energy (290.10 MeV) serves as an explicit, quantitative gravitational source.

The two tracks converge: both derive the same equation from the same GTE substrate, at different levels of description. Their unification requires the CMCA-to-Φ_{MDL} continuum limit (machine-certified in P42) and the identification of the discrete Ollivier–Ricci curvature with the coarse-grained Ricci tensor of Φ_{MDL} (an open problem, the subject of future work).

The geodesic interpretation of particle motion complements the stress-energy derivation of this paper. In P36 [15], the Rule 110 τ_c mechanism was shown experimentally to select geodesic paths (minimum computational effort): the τ_c ratio matches γ_{SR} at $p = 4.2 \times 10^{-73}$ and the equivalence principle is confirmed at $p = 8.9 \times 10^{-14}$. The continuum Φ_{MDL} geodesic theorem (`tau_c_prefers_geodesic`, `psc_orbit_is_curvature_geodesic`, $\mathbf{A}_{\text{Lean}}$, zero `sorry`; certified in P43 [12]) certifies that this experimental result holds as a formal theorem: particles follow geodesics of minimum τ_c effort, consistent with the Einstein equations derived here.

8.2 Relation to existing approaches

Causal dynamical triangulations (CDT) [1] and spin-foam models [9] derive Einstein gravity from discrete combinatorial structures by taking a suitable continuum limit. The GTE approach differs in two respects. First, the underlying discrete structure (Rule 110 CA) is uniquely selected by the Standard Model orbit structure, rather than introduced ad hoc. Second, both matter content (SM generations, quark confinement) and spacetime geometry (spectral dimension $d_s = 4$, Ollivier–Ricci curvature) emerge from the same arithmetic rule, without separate matter and gravity sectors.

The Gorard [4] result on Ollivier–Ricci curvature in Wolfram-model causal graphs provides the closest precedent for the CA-track result in P36. The present Φ_{MDL} track is independent: it does not require the causal graph and derives the Einstein equations at the continuum field-theory level.

Jacobson's [6] thermodynamic derivation of the Einstein equation from the Clausius relation $\delta Q = T\,dS$ on local Rindler horizons provides another independent route. The GTE MDL-Lovelock correspondence provides yet a third structural route—from information-theoretic minimality rather than thermodynamics. All three routes yield the same equation, supporting the universality of Einstein gravity as the low-energy effective theory.

8.3 Relation to P37 (quantum mechanics) and P42 (Φ_{MDL} field)

Paper P37 [20] develops the quantum-mechanical sector over the Φ_{MDL} kink modes: Fock quantization, the Born rule as a closure fixed point, and 't Hooft's cogwheel construction. Paper P42 [18] establishes Poincaré invariance and the Born rule for Φ_{MDL} at $\mathbf{A}_{\mathrm{Lean}}$. The present paper provides the gravitational coupling: Φ_{MDL} kink states are not only quantum excitations (P37) but also gravitational sources (P38). The two sectors—quantum and gravitational—arise from the same field Φ_{MDL}, consistent with the beable-level unification proved in `QuantumGravity.lean`.

9 Gravitational Hierarchy: Derivation of Newton's Constant

This section presents the GTE derivation of Newton's constant G_N: a Lean-certified identity connecting the QCD β coefficient to the GTE substrate, and an orbit-decomposition theorem with Z_3-invariant entropy mechanism yielding a $\mathbf{A/D}$ derivation of the Planck-scale hierarchy.

9.1 The $b_0 = |Z_7|$ identity

The one-loop QCD β-function coefficient is $b_0 = (11N_c - 2N_f)/3$. With $N_c = 3$ colours and $N_f = 6$ active flavours (three generations, $N_f = 2N_{\mathrm{gen}} = 6$):

$$b_0 = \frac{11 \cdot 3 - 2 \cdot 6}{3} = \frac{33 - 12}{3} = \frac{21}{3} = 7.$$

The GTE substrate contains the Frobenius group $F_{21} = Z_7 \rtimes Z_3$ of order $|F_{21}| = 21$, whose unique normal Z_7 subgroup has order $|Z_7| = 7$. Writing $11N_c = 11|Z_3| = 33$ and $2N_f = 2 \cdot 2N_{\mathrm{gen}} = 12$, we obtain

$$b_0 = \frac{|F_{21}|}{|Z_3|} = |Z_7| = 7.$$

The identity $b_0 = |Z_7|$ is Lean-certified (`b0_eq_z7_order`, `BetaCoefficientIdentity.lean`, zero `sorry`): the QCD asymptotic-freedom coefficient is not a free parameter but is fixed by the group-theoretic structure of the GTE substrate.

9.2 Orbit decomposition of the CMCA 3-cell neighborhood space

The CMCA rule operates on 3-cell neighborhoods drawn from Z_7^3 (343 states). The group F_{21} acts diagonally on Z_7^3 by $(a, s) \cdot x = 2^s x + a$ (componentwise). By Burnside's lemma, the action has $(7^3 + 14)/21 = 17$ orbits. These decompose by neighborhood type:

Type	States	Orbit size	# Orbits
All-equal (k, k, k)	7	7	1
Exactly-two-equal	126	21	6
All-distinct	210	21	10
Total	343	—	17

The all-distinct count follows from $|Z_7|(|Z_7| - 1)(|Z_7| - 2)/|F_{21}| = 210/21 = 10$ orbits; the degenerate count $(1 + 6 = 7 = b_0 = |Z_7|)$ is a consequence of the $b_0 = |Z_7|$ identity: the number of degenerate F_{21}-orbits on Z_7^3 equals the QCD one-loop β coefficient. This identity is Lean-certified (`gte_planck_cascade_group_identity`, `BetaCoefficientIdentity.lean`).

Let $n = 10$ denote the number of all-distinct orbits (the generic sector) and $b_0 = 7$ the number of degenerate orbits (the degenerate sector). The total $n + b_0 = 17$ agrees with the Burnside count.

9.3 The Planck-hierarchy formula

The ratio $M_{\rm Pl}/m_\tau = (1.220 \times 10^{22}\ \text{MeV})/(1776.86\ \text{MeV}) = 6.871 \times 10^{18}$ satisfies:

$$\frac{M_{\rm Pl}}{m_\tau} \approx \frac{|F_{21}|^n\,|Z_7|^{b_0}}{2} = \frac{21^{10} \times 7^7}{2},$$

with a relative error of 0.040% against the PDG values. Equivalently, writing the ratio as an entropy:

$$\ln\!\left(\frac{M_{\rm Pl}}{m_\tau}\right) = n \ln|F_{21}| + b_0 \ln|Z_7| - \ln 2,$$

where the exponents count the generic $(n = 10)$ and degenerate $(b_0 = 7)$ orbit types in the GTE CMCA neighborhood space. Every parameter in this expression is an independently certified GTE quantity: $|F_{21}| = 21$ and $|Z_7| = 7$ are the orders of the GTE substrate and its normal subgroup; $n = 10$ is the number of all-distinct F_{21}-orbits on Z_7^3; $b_0 = 7 = |Z_7|$ is the QCD one-loop coefficient (GTE theorem above). There are no free parameters.

The formula admits an MDL interpretation: the Planck mass is the UV cutoff of the GTE configuration space, at which every orbit parameter is specified. The generic-sector orbits contribute a factor $|F_{21}|$ each (full F_{21}-orbit size, trivial stabilizer), and the degenerate-sector orbits contribute $|Z_7|$ each (reduced size due to non-trivial Z_3 stabilizer of the all-equal diagonal); the factor of 2 arises from the outer Z_2 binary sublayer of the CMCA.

Subsequent analysis derived both the exponent n and the denominator from GTE first principles. The exponent $n = 10$ counts the orbit-specific neighborhoods in the CMCA `fmdl` lookup table: $n = (N_{\rm gen} - 1) \times N_{\rm cells} = (3 - 1) \times 5 = 10$, where $N_{\rm gen} = 3$ (Lean-certified) and $N_{\rm cells} = 5$ (the PSC beable dimension). The denominator $|Z_2| = 2$ counts the exactly two PSC-admissible CMCA variants: Rule 110 and Rule 124, which preserve the PSC orbit identically but differ on precisely the two chiral-conjugate neighborhoods $(0, 0, 1) \leftrightarrow (1, 0, 0)$, related by Z_2 space-reflection. With these derivations, all parameters in the formula are GTE-derived.

The two independent derivations of $n = 10$ are provably the same theorem. The unifying identity is the **GTE Frobenius prime condition**: $|Z_7| = |Z_3|^2 - |Z_3| + 1 = 9 - 3 + 1 = 7$, which connects F_{21} orbit theory (group theory) to PSC beable dynamics (cellular automaton). Formally: $(|Z_7|-1)(|Z_7|-2)/|Z_3| = (N_{\rm gen}-1)(|Z_7|-2)$ iff $|Z_7| = |Z_3|^2 - |Z_3| + 1$. This identity is Lean-certified in `FrobeniusPrimeIdentity.lean` (zero `sorry`). The $M_{\rm Pl}/m_\tau$ formula is **A/D**: all components and the orbit-weight mechanism are derived from GTE first principles.

Proposition 9.1 (Z_3-invariant entropy theorem, **A/D** derivation with one open Lean step (disjointness lemma); $\mathbf{A}_{\rm Lean}$ cert in P43)**.** *For the diagonal F_{21}-action on Z_7^3, the MDL weight assigned to each orbit type in $S_{\rm GTE}$ is:*

- All-distinct orbits *(size 21, trivial stabiliser, PSC-selected): weight* $\ln|F_{21}| = \ln 21$*.*
- Two-equal orbits *(size 21, trivial stabiliser, PSC-unselected): decompose into exactly three Z_3-conjugate Z_7-translation sub-orbits of size 7; Z_3 acts freely and transitively. In the Z_3-invariant (colour-neutral) sector the effective weight is* $\ln|Z_7| = \ln 7$*.*
- The all-equal orbit *(size 7, stabiliser Z_3): weight* $\ln|Z_7| = \ln 7$ *by the orbit-stabiliser theorem.*

Proof sketch: *if $(a+k, a+k, b+k) = (2a+j, 2a+j, 2b+j)$ for $a \neq b$, then $k-j \equiv a$ and $k-j \equiv b$, a contradiction; so $T(s) \cap b \cdot T(s) = \varnothing$. All six two-equal orbits verified computationally.*

Proposition 9.1 closes the orbit-weight mechanism, deriving Newton's constant G_N from GTE at **A/D**. The $\mathbf{A}_{\rm Lean}$ Lean certification is provided in P43 [12] (`z3_invariant_entropy_closes_hierarchy` in `Z3InvariantEntropy.lean`, zero `sorry`).

Remark 9.2 (Gravitational hierarchy orbit-weight mechanism — **A/D** here; $\mathbf{A}_{\rm Lean}$ in P43)*.* The orbit-weight mechanism is closed by Proposition 9.1 above. All components of the formula are GTE-derived, constituting a **A/D** derivation of Newton's constant G_N. The $\mathbf{A}_{\rm Lean}$ Lean certification (`z3_invariant_entropy_closes_hierarchy`, `Z3InvariantEntropy.lean`, zero `sorry`) is established in P43 [12].

10 Conclusions

We have derived the stress-energy tensor $T_{\mu\nu}$ of the $\mathbb{Z}_7$-symmetric Klein–Gordon field $\Phi_{\rm MDL}$ analytically and certified key properties in Lean 4 (zero `sorry`). Combined with the MDL-Lovelock structural correspondence established in P35, we obtain the linearized Einstein field equations $G_{\mu\nu} = 8\pi G_N T_{\mu\nu}[\Phi_{\rm MDL}]$ by standard variational calculus.

The BPS kink is a well-defined gravitational source with rest mass 290.10 MeV (relative error 1.4×10^{-6}), pressure-free longitudinal stress ($T_{11} = 0$ to floating-point precision), and transverse tension $T_{22} = T_{33} = -T_{00}$. Energy-momentum conservation holds to 6×10^{-12} numerically.

The classical cosmological constant vanishes identically at all seven $\mathbb{Z}_7$ vacua ($V(\Phi_k) = 0$ exactly, Lean-certified), but the one-loop correction $|\Delta V_{\rm CW}| \approx 1.7 \times 10^7$ MeV^4 exceeds the observed dark-energy density by $\sim 10^{42}$—the cosmological constant problem in the GTE context, identified but unresolved.

The quantum gravity scale is $M_{\rm Pl}^{\rm GTE} = \pi/\sqrt{3} \approx 1.81$ lattice units, at which the Nyquist residual $\varepsilon_0(M) = \pi^2/(3M^2)$ reaches $O(1)$ and the classical spacetime approximation breaks down. A single GTE kink decoheres via Penrose objective reduction on a timescale of ~ 20 million years.

Newton's constant G_N is derived from GTE first principles at **A/D** via the gravitational hierarchy formula (Proposition 9.1).

A Lean Inventory

Table 4 summarizes all Lean 4 certifications relevant to this paper. All theorems are in the `ugp-lean` repository (public); zero `sorry` unless noted.

Table 4: Lean 4 theorem inventory for P38.

Theorem / Lean name	**Status**	**Content**
`phimdl_tmunu_symmetric`	$\mathbf{A}_{\text{Lean}}$	$T_{\mu\nu} = T_{\nu\mu}$; `StressEnergyTensor.lean`.
`phimdl_potential_at_vacuum_zero`	$\mathbf{A}_{\text{Lean}}$	$V(\Phi_k) = 0$ at $\mathbb{Z}_7$ vacuum; `StressEnergyTensor.lean`.
`phimdl_tmunu_vacuum_zero`	$\mathbf{A}_{\text{Lean}}$	$T_{\mu\nu}\|_{\text{vac}} = 0$; `StressEnergyTensor.lean`.
`phimdl_bps_kink_pressure_free`	axiom	$T_{11} = 0$ on BPS kink; BPS calculus pending formalization. Numerically verified to 5×10^{-17}.
`phimdl_gravity_sector_prerequisites`	$\mathbf{A}_{\text{Lean}}$	Bundle: Poincaré inv. + sym. + vac-zero + no-CA-replica; `StressEnergyTensor.lean`. One axiom (BPS slot).
`gte_is_beable_level_quantum_gravity`	**A/D**	Geometry + particles + dynamics from f_{MDL}; `QuantumGravity.lean`. D2 sub-components $\mathbf{A}_{\text{Lean}}$.
`matter_geometry_from_same_rule`	**A/D**	No separate matter/geometry sectors; `QuantumGravity.lean`.
`particles_source_and_follow_curvature`	**A/D**	Gliders source $\kappa_{\text{SD}} > 0$ and follow causal geodesics; `QuantumGravity.lean`.
`unification_is_not_bolt_on`	**A**	Three sectors forced by f_{MDL}; `QuantumGravity.lean`.
`async_algebraic_lifting_theorem`	$\mathbf{A}_{\text{Lean}}$	Async update preserves PSC-admissibility; `AsyncLiftingTheorem.lean`, zero axioms.
`async_color_confinement`	$\mathbf{A}_{\text{Lean}}$	Color confinement under async update; derived from above.
`causal_graph_spectral_dim_thermodynamic_limit`	$\mathbf{A}_{\text{Lean}}$	$d_s = 4$ in thermodynamic limit; P36 / `SpectralDimension.lean`.
`poincare_invariance_of_kg`	$\mathbf{A}_{\text{Lean}}$	KG dispersion $\omega^2 = k^2 + m^2$; P42 / `LorentzInvariance.lean`.

(continued on next page)

(Table 4 continued)

Theorem / Lean name	Status	Where / content
`b0_eq_z7_order`	$\mathbf{A}_{\text{Lean}}$	$b_0 = (11N_c - 2N_f)/3 = \lvert F_{21}\rvert/\lvert Z_3\rvert = \lvert Z_7\rvert = 7$; zero sorry; `BetaCoefficientIdentity.lean`.
`gte_planck_cascade_group_identity`	$\mathbf{A}_{\text{Lean}}$	Degenerate F_{21}-orbit count on Z_7^3 equals $b_0 = \lvert Z_7\rvert$; generic-orbit count $n = 10$; $n + b_0 = 17$ (Burnside total); `BetaCoefficientIdentity.lean`.
`gte_newtonian_force_law_continuum`	**A/D**	$F(b) = G_{\text{eff}} M/(4\pi b^2)\,[1 - \sigma_{\text{AL}}^2/(2b^2) + \cdots]$ as $b/\sigma_{\text{AL}} \to \infty$; `PMDLGravityTheorems.lean`. Full $\mathbf{A}_{\text{Lean}}$ requires Mathlib 3D Poisson kernel (forthcoming).
`tau_c_three_mechanisms_equivalent`	**A/D**	Gradient-kick (Level 1), linearized GR metric perturbation (Level 2), and beable equivalence principle (Level 3) give the same Newtonian force; `PMDLGravityTheorems.lean` (structural).

B Computational Methods

Four Python scripts produced the numerical results reported in this paper. All scripts are available in `papers/38_emergent_gravity_phimdl/scripts/`.

`phimdl_tmunu_full.py`. Computes the full $T_{\mu\nu}$ tensor for the Φ_{MDL} BPS kink. Uses $m_\varphi = 1776.86$ MeV, kink profile $\Phi_{\text{kink}}(x) = (4/7)\arctan(\exp(m_\varphi x))$ with $N = 1000$ grid points over $[-20/m_\varphi,\ 20/m_\varphi]$. Verifies BPS saturation ($\lvert\partial_x\Phi - \sqrt{2V}\rvert_\infty < 2.8 \times 10^{-5}$), $\lvert T_{11}\rvert_\infty < 10^{-16}$, $\lvert T_{00} + T_{22}\rvert_\infty < 10^{-16}$, $\int T_{00}\,dx = 290.10$ MeV (midpoint rule, $N = 1000$), moving-kink Lorentz test at six velocities, and conservation $\lvert\partial_\mu T^{\mu\nu}\rvert_\infty < 6.2 \times 10^{-12}$.

`phimdl_kink_gravitational_source.py`. Computes the gravitational-source properties of the BPS kink. Uses 100,000-point midpoint integration for $\int T_{00}\,dx$. Computes FWHM, Fourier form factor via DFT, Schwarzschild radius, and gravitational self-energy ratio $\lvert U_{\text{grav}}\rvert/M_{\text{kink}} \approx 3.4 \times 10^{-39}$.

`phimdl_cosmological_constant.py`. Evaluates $V(\Phi_k)$ at all seven $\mathbb{Z}_7$ vacua, the potential barrier $V_{\max} = 2m_\varphi^2/49$, the one-loop Coleman–Weinberg correction $\Delta V_{\text{CW}} = -3m_\varphi^4/(128\pi^2)$, and the hierarchy ratio $\lvert\Delta V_{\text{CW}}\rvert/\rho_\Lambda^{\text{obs}}$.

`phimdl_quantum_gravity_regime.py`. Computes the Nyquist residual $\varepsilon_0(M) = \pi^2/(3M^2)$ over $M \in [1,\ 1000]$, identifies the threshold $M_{\text{Pl}}^{\text{GTE}} = \pi/\sqrt{3}$, and computes the Penrose–Diósi decoherence rate $\Gamma_{\text{OR}} = G_N M_{\text{kink}}^2 m_\varphi/\hbar^2 c$ and the corresponding OR time $\tau_{\text{OR}} \approx 20$ Myr.

All scripts use `numpy` for numerical integration and standard physical constants from CODATA/PDG. Results are stored as JSON artifacts co-located with the scripts. Dependencies: Python 3.9+, numpy.

Acknowledgments

The Lean 4 certifications in this paper were developed using the Lean theorem prover and Mathlib library. Numerical computations were performed using Python and numpy.

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